# Federated Learning for Skin Lesion Classification under Natural Source Heterogeneity: A Lesion-Safe Evaluation on HAM10000

Research Proposal

Nguyen Xuan Binh (SE204049)
*Group 9 - AI2007*
*FPT University*
ngxbinh9999@gmail.com

Pham Vu Gia Kiet (SE203905)
*Group 9 - AI2007*
*FPT University*
gkpham2006@gmail.com

Nguyen Bao Dang (SE203688)
*Group 9 - AI2007*
*FPT University*
nguyenbaodang3@gmail.com

***Abstract*—Dermoscopic images are collected by different clinics with different acquisition devices, and privacy rules often prevent these images from being pooled. Federated learning (FL) trains a shared model without moving images, but published FL studies on skin lesion data frequently simulate clients with artificial partitions and rarely state whether images of the same lesion are kept on one side of the train/test split. This proposal uses the four acquisition sources recorded in HAM10000 as four naturally heterogeneous clients and evaluates FL under a lesion-level split. HAM10000 contains 10,015 images of 7,470 lesions, and its sources differ strongly in class composition (one source is 94.1% nevi, another holds 90.2% of all AKIEC images). We ask how far FedAvg falls below centralized training in Macro-F1 under this lesion-safe, source-based setting, whether the gap concentrates in classes held mainly by one client, and whether FedProx narrows it, with a FedBN-style variant as a conditional extension. Four arms (centralized, local-only, FedAvg, FedProx) share one backbone and one training budget and are compared on a pooled test set. The proposal reports the dataset audit and partition already completed and the experimental plan.**

***Index Terms*—federated learning, skin lesion classification, HAM10000, non-IID data, lesion-level split, evaluation protocol**

## I. Introduction

Deep learning classifiers for dermoscopic images need large and diverse training sets, but such images are distributed across hospitals and clinics that use different imaging devices and serve different patient populations. Sharing the raw images is often restricted by privacy regulations and institutional policy. Federated learning (FL) addresses this setting by keeping data at each site and exchanging only model parameters: a server sends a global model to the clients, each client trains locally, and the server aggregates the updates [1]. When the clients are a small number of institutions with medium to large datasets, the setting is called cross-silo FL [2].

The main difficulty of FL in medicine is statistical heterogeneity: clients differ in data volume, class composition and image appearance. Many FL studies simulate this heterogeneity by artificially splitting a pooled dataset, for example with Dirichlet sampling on the labels. Benchmark work has argued that such synthetic partitions may not reproduce the heterogeneity of real multi-site data and that natural client boundaries are a safer basis for evaluation [2].

HAM10000 [3] offers such natural boundaries. Its metadata records the acquisition source of each image, and the four sources differ strongly in class composition (Section III). HAM10000 also contains several images of the same lesion: 1,956 of its 7,470 lesions have at least two images. In centralized studies, splitting at image level instead of lesion level has been shown to change reported performance substantially [4]. In FL, a split performed independently inside each client can additionally place images of the same lesion in different clients and different splits. Among the federated studies we screened on dermoscopy data, the train/test split is described at image level or not described at lesion level [2], [5].

This proposal therefore studies FL on HAM10000 with two controls applied together: clients defined by the real acquisition source, and a train/validation/test split performed by `lesion_id` on the whole dataset before images are assigned to clients. The central question is how much performance is lost when the same model is trained federatedly rather than on pooled data, once both of these controls are in place. Specifically, we measure how far the Macro-F1 of FedAvg falls below that of centralized training when both use the same backbone and the same training budget. We then ask where this loss comes from: whether it is concentrated in classes whose training images are held mainly by one client, such as AKIEC, of which 90.2% comes from a single source. Finally, we examine whether FedProx [6], which limits how far each client's model drifts from the global model, narrows the gap at the same budget. A FedBN-style variant that keeps BatchNorm layers local [7] is considered only as a conditional extension (Section IV-D).

These questions are to be tested experimentally. The planned contributions are: (i) a lesion-safe, source-based federated partition of HAM10000 that is released with fingerprints for reproducibility; (ii) a controlled comparison of centralized, local-only, FedAvg and FedProx training under one budget and one pooled-test metric; and (iii) a per-class analysis that links

the FL gap to how classes are distributed across clients. The project does not implement encryption or differential privacy and makes no formal privacy claim.

## II. Related Work

### A. Centralized classification on HAM10000

HAM10000 [3] served as the training set of the ISIC 2018 classification challenge [8] and remains a common benchmark. Recent centralized studies cover Transformer architectures [9], hybrid CNN–LSTM models in a binary setting [10], and GAN-based augmentation for class imbalance [11]. These studies report high headline scores, but lesion-level grouping is not reported in their splits. Two recent studies address evaluation directly: Yu [12] uses an explicit lesion-level split and reports a Macro-F1 of 0.625 for an image-only ResNet50, and Jamaludin and Kim [4] show that correcting a leakage-prone split to a verified lesion-level split lowers accuracy from 94.57% to 78.41% for the same pipeline. Evaluation protocol is therefore a first-order factor when HAM10000 results are compared.

### B. Federated optimization

FedAvg [1] trains a shared model by iterative model averaging: clients run local updates and the server averages their parameters weighted by local sample counts. It is robust to moderately non-IID data but can become unstable when clients differ strongly and run many local steps. FedProx [6] adds a proximal term to each local objective that penalizes the distance between the local and global parameters. With the penalty set to zero it reduces to FedAvg. FedBN [7] targets feature-shift heterogeneity, such as images from different scanners, by excluding BatchNorm parameters from aggregation so that each client keeps its own normalization layers. Because FedBN produces client-specific models, its evaluation is reported on each client's own test set.

### C. Federated learning on dermoscopy data

FLamby [2] provides cross-silo healthcare benchmarks with natural client splits. Its Fed-ISIC2019 task partitions 23,247 ISIC 2019 images into six clients by acquisition system. Four of these clients are the HAM10000 sources used in this proposal, with identical image counts (3,954, 3,363, 2,259 and 439). On Fed-ISIC2019, FL strategies improve over local training but do not reach pooled (or centralized) training. FLamby reports balanced accuracy averaged uniformly over the clients' test sets, and its per-client train/test split is not described at lesion level. Hendrix et al. [5] also partition Fed-ISIC2019 by source and compare FedAvg, FedProx and prompt-based methods on a binary nevus-versus-other task. They split each client's data 75/25 at image level, report per-client balanced accuracy, and include no centralized reference.

### D. Gap addressed by this proposal

Table I compares the evaluation settings. Among the studies we screened, natural source-based clients, a lesion-level split, a centralized reference and a pooled-test metric are not combined in one evaluation of HAM10000. This proposal combines them so that the FL gap can be measured without lesion leakage and compared across classes.

TABLE I
Evaluation settings of screened FL studies on dermoscopy data and of this proposal

| | FLamby [2] | Hendrix [5] | This proposal |
|---|---|---|---|
| Data | ISIC 2019, 8 classes | ISIC 2019, binary | HAM10000, 7 classes |
| Clients | 6, by source | 6, by source | 4, by source |
| Train/test split | Per client; level not stated | Per client, image level | Global, lesion level |
| Centralized reference | Yes | No | Yes |
| Metric | Balanced acc., mean over clients | Balanced acc., per client | Macro-F1, pooled test |

## III. Dataset and Federated Partition

### A. HAM10000

We use the HAM10000 collection [3], [13], distributed as the ISIC 2018 Task 3 training set [8]. It contains 10,015 RGB dermoscopic images of 7,470 lesions in seven classes: melanocytic nevi (NV, 6,705), melanoma (MEL, 1,113), benign keratosis-like lesions (BKL, 1,099), basal cell carcinoma (BCC, 514), actinic keratoses/intraepithelial carcinoma (AKIEC, 327), vascular lesions (VASC, 142) and dermatofibroma (DF, 115). NV is about 58 times more frequent than DF. The task is 7-class classification of a single image.

### B. Lesion-level split

Because 1,956 lesions have two or more images, images are not independent observations. We reuse the frozen lesion-level split from our dataset audit: the 7,470 lesions are split 70/15/15, stratified by diagnosis, with seed 42 and no post-hoc adjustment. This yields 6,981 / 1,532 / 1,502 images (5,229 / 1,120 / 1,121 lesions) for train / validation / test, with zero lesion overlap between any two splits. The split manifest is fingerprinted.

### C. Source-based clients

The metadata column `dataset` records four acquisition sources, which we use as four clients: `vidir_molemax`, `vidir_modern`, `rosendahl` and `vienna_dias`. Table II lists their class counts and Fig. 1 shows the class proportions within each client. The clients are strongly non-IID: `vidir_molemax` is 94.1% NV, `rosendahl` holds 295 of the 327 AKIEC images (90.2%), and the two Vienna sources contain no AKIEC. Client sizes are also unbalanced, from 3,954 images (39.5%) to 439 images (4.4%). The lesion structure differs as well: `vidir_molemax` has exactly one image per lesion, while the other sources contain multiple images per lesion.

TABLE II
CLASS DISTRIBUTION OF THE FOUR HAM10000 ACQUISITION SOURCES USED AS FEDERATED CLIENTS

| Client | Images | NV | MEL | BKL | BCC | AKIEC | VASC | DF |
|---|---|---|---|---|---|---|---|---|
| vidir_molemax | 3,954 | 3,720 | 24 | 124 | 2 | 0 | 54 | 30 |
| vidir_modern | 3,363 | 1,832 | 680 | 475 | 211 | 32 | 82 | 51 |
| rosendahl | 2,259 | 803 | 342 | 490 | 296 | 295 | 3 | 30 |
| vienna_dias | 439 | 350 | 67 | 10 | 5 | 0 | 3 | 4 |

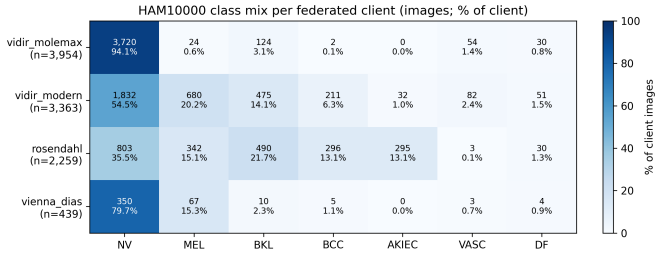



Fig. 1. Class proportions within each of the four HAM10000 sources used as clients. Proportions are computed within a client. The concentration of AKIEC in `rosendahl` is stated in the text.

### D. Leakage-safe assignment

Nine lesions (26 images, all MEL) appear in both `vidir_modern` and `vidir_molemax`. If the data were split separately inside each client, the same lesion could fall into the training set of one client and the test set of another. We therefore apply the global lesion-level split first and only then assign images to clients by their source (Fig. 2). All nine cross-source lesions lie entirely within one split. The resulting partition is fingerprinted. Within each client, the training share ranges from 68.7% to 70.6%, so the global split is not biased toward any source.

### E. Excluded metadata

Two metadata columns are not used as model inputs. All 3,704 images with `dx_type` = follow-up are NV, so this column nearly reveals the label. The `dataset` column defines the clients and is used only for partitioning and analysis. The model receives the image only.

### F. Consequences for evaluation

In the training split, `vidir_molemax` has no BCC or AKIEC images and `vienna_dias` has no AKIEC images, so some clients never observe some classes. The per-client validation and test sets are small: 26 of the 56 client-split-class cells (4 clients $\times$ {val, test} $\times$ 7 classes) contain fewer than ten images, and the validation set of `vienna_dias` contains a single MEL image. Per-client metrics would therefore be unstable. The primary metric is computed on the pooled test set (1,502 images), and per-client results are reported only as secondary analysis.

## IV. METHODOLOGY

### A. Training arms

Four arms are compared. All use the same backbone, input resolution, augmentation, loss and total training budget.

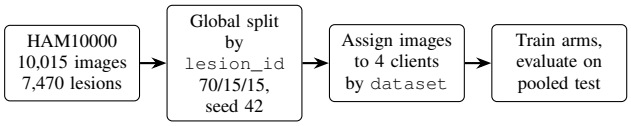



Fig. 2. Lesion-safe federated partition. The split is fixed before client assignment, so no lesion crosses splits, including the nine lesions shared by two sources.

- Centralized (upper reference): one model trained on the pooled training split of all four clients.
- Local-only (lower reference): one model per client, trained only on that client's training data with no collaboration.
- FedAvg: federated training with sample-weighted averaging [1].
- FedProx: FedAvg with a proximal term in each local objective [6].

All four clients participate in every round. Clients are simulated sequentially on one machine so no network communication is involved.

### B. FedAvg

Let $K = 4$ be the number of clients, $n_k$ the number of training images of client $k$ and $n = \sum_k n_k$. In round $t$ the server sends the global parameters $w^t$ to every client. Client $k$ initializes from $w^t$, performs $E$ local epochs of stochastic gradient descent on its loss $F_k$, and returns $w_k^{t+1}$. The server then sets

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1}. \tag{1}$$

With this weighting, `vidir_molemax` contributes about nine times more than `vienna_dias`. In plain FedAvg, BatchNorm parameters and running statistics are averaged in the same way as the other parameters.

### C. FedProx

FedProx [6] replaces each local objective with

$$\min_{w} \; F_k(w) + \frac{\mu}{2} \|w - w^t\|^2, \tag{2}$$

where $\mu \geq 0$ controls how far the local model may move from the current global model. With $\mu = 0$, FedProx is identical to FedAvg. Aggregation follows (1). The value of $\mu$ is selected on the pooled validation set from a small grid.

### D. FedBN-style extension (conditional)

FedBN [7] keeps BatchNorm layers local and averages only the remaining parameters. The three Vienna sources use different imaging systems, so feature shift is plausible in our data. However, FedBN yields a client-specific model for each client and, as in the original study, is evaluated on each client's own test set. This does not match our pooled-test primary metric, and several per-client test cells are very small. FedBN is therefore not part of the main comparison. If time allows, it will be run as an extension and reported with per-client evaluation as secondary analysis. The decision is recorded as open in our project log.

### E. Backbone and training setup

Two ImageNet-pretrained backbones are candidates: ResNet18 and EfficientNet-B0. Exactly one becomes the primary backbone for all arms and the other is used only for a robustness check (Section V). Images are resized to $224 \times 224$. Training-related hyperparameters (loss, optimizer, learning rate, batch size, augmentation) are tuned once for centralized training on the pooled training and validation data and then frozen for all arms, following the practice of FLamby [2]. Only FL-specific parameters (local epochs $E$, rounds $R$, $\mu$) are set separately. Experiments use PyTorch on a single NVIDIA RTX 5070 GPU (12 GB).

### F. Evaluation

The primary metric is Macro-F1, the unweighted mean of the seven per-class F1 scores, computed on the pooled test set. Accuracy is not used as the primary metric because NV is 67% of the data and a model can reach high accuracy while missing DF, VASC and AKIEC. Secondary metrics are per-class recall, balanced accuracy, accuracy and the confusion matrix. For FedAvg and FedProx the final global model is evaluated while for local-only training each client's model is evaluated on the same pooled test set. Model selection (best round or epoch) uses the pooled validation set only, and the test set is used once per final model. Uncertainty is reported with bootstrap confidence intervals that resample test lesions rather than images, so that images of the same lesion stay together.

## V. Experimental Plan

### A. Hypotheses

We expect that, at equal training budget, FedAvg reaches a lower pooled-test Macro-F1 than centralized training but a higher Macro-F1 than local-only training. FLamby reports this ordering on Fed-ISIC2019 with balanced accuracy [2]. We test whether it holds, and how large the gap is, under a lesion-level split and Macro-F1. We further expect the per-class recall drop from centralized training to FedAvg to be largest for classes that are concentrated in one client or absent from some clients' training data, notably AKIEC and BCC. Lastly, we expect FedProx with a tuned $\mu > 0$ to reduce the gap between FedAvg and centralized training at the same budget. Each expectation is stated so that it can be rejected by the results.

### B. Variables

Table III summarizes the design. The independent variable is the training arm, everything else is held fixed.

### C. Training budget

To compare arms fairly, the federated budget is matched to centralized training: with $R$ rounds of $E$ local epochs, each client passes over its data about $R \times E$ times, which is set equal to the number of centralized epochs. This mirrors the budget rule used in FLamby [2].

TABLE III
Experimental variables

| | |
|---|---|
| Independent | Training arm: centralized, local-only, FedAvg, FedProx |
| Fixed | Lesion-level split (seed 42); 4 source-based clients; backbone; $224 \times 224$ input; augmentation; loss; optimizer; total budget |
| FL-specific | Local epochs $E$, rounds $R$, proximal weight $\mu$ |
| Outcome | Pooled-test Macro-F1 (primary); per-class recall, balanced accuracy, accuracy (secondary) |

### D. Backbone selection rule

The rule is fixed before any run. ResNet18 and EfficientNet-B0 are trained centrally under identical settings and compared on pooled validation Macro-F1 only. The backbone with the higher score becomes primary: if the difference is below 0.02, ResNet18 is chosen because it is simpler to train. The test set is not used, and no FL run is involved, so the choice cannot be tuned to make the FL gap look larger or smaller. Training time per epoch and peak GPU memory are recorded to plan the FL runs.

### E. Robustness check

If time allows, the centralized and FedAvg arms are repeated with the secondary backbone. The question is whether the direction and approximate size of the FL gap are the same for both backbones. The primary backbone is not changed after this check. Main arms are repeated with three random seeds when compute permits.

### F. Decision points and fallbacks

- If the PyTorch training loop is not running end-to-end by the end of Week 6, Week 7 is spent on the centralized baseline only, and FL work starts after it is stable.
- If FedAvg does not converge (Macro-F1 does not improve over rounds), the number of local epochs is reduced. The centralized versus local-only comparison remains reportable.
- If FedAvg is close to centralized training, so that source-based heterogeneity alone produces no clear gap, a Dirichlet label partition is added as a secondary analysis to vary the degree of heterogeneity.

## VI. Timeline and Demo Plan

The proposal is presented at the end of week 5 of the semester, and the final report is presented in the last class session (estimated). Table IV lists the work planned for weeks 6 -10. If there are any intermediate milestones announced, the plan will be aligned with them.

If the federated runs are delayed, Week 9 is reserved for completing FedAvg and FedProx, and the robustness check and FedBN extension are dropped first.

A Gradio web application will let a user upload a dermoscopic image and see the predicted class probabilities of the federated model next to those of the centralized model. The demo is for illustration only and is not a diagnostic tool.

TABLE IV
PLANNED TIMELINE

| Week | Task | Output |
|---|---|---|
| 6 | Incorporate proposal feedback; verify PyTorch on the GPU; data loader for the partition file; backbone selection pilot (Section V) | Running pipeline; primary backbone |
| 7 | Centralized baseline; local-only training for each client | Upper and lower references |
| 8 | FedAvg, then FedProx with $\mu$ grid, same budget | FL results |
| 9 | Per-class analysis of the FL gap; robustness check with the secondary backbone; optional FedBN extension; demo application | Analysis tables and figures; demo |
| 10 | Final report, slides and reproducibility package; final presentation | Report, code, logs |

## VII. LIMITATIONS

The study is a simulation. All four clients run on one machine, so communication cost, client dropout and differences in client hardware are not studied, and model selection relies on a pooled validation set held by the server, which would not exist in a real federation. With only four clients, one of which (`vienna_dias`) has 439 images, the conclusions may not transfer to federations with many more sites.

The clients also differ in ways that go beyond class proportions. Some have no training images of certain classes (BCC and AKIEC in `vidir_molemax`, AKIEC in `vienna_dias`), which is part of the heterogeneity we study but also limits what local-only models can learn. In addition, `vidir_molemax` contains one image per lesion while the other clients contain several, so the effective number of independent samples is not proportional to the image counts.

Finally, the scope of the claims is limited. HAM10000 comes from two institutions and does not represent the clinical population, so the results are not clinical evidence. They are also not directly comparable with FLamby or other FedISIC2019 studies, which use eight classes, balanced accuracy averaged over clients, and splits not described at lesion level. This study do not use encryption, secure aggregation or differential privacy: raw images remain on the clients, but no formal privacy guarantee is claimed.

## VIII. REPRODUCIBILITY AND DATA LICENSE

### A. Data and partition

All experiments load two frozen files: the lesion-level split manifest (SHA-256 prefix `bda53d19`) and the federated partition file listing `image_id`, `lesion_id`, label, split and client (prefix `0ab49e23`). Each script checks these fingerprints before training and stops if they differ. Every dataset number in this proposal is computed by code from these files and the original HAM10000 metadata.

### B. Experiments

For each run we record the backbone, arm, random seed, $R$, $E$, $\mu$, learning rate, batch size, number of epochs, selected round or epoch, validation and test metrics, training time and peak GPU memory, together with library versions. Configuration files, training logs, evaluation scripts and the demo will be released with the final report.

### C. License

HAM10000 (Harvard Dataverse, version 4) is distributed under the Creative Commons Attribution-NonCommercial 4.0 International license (CC BY-NC 4.0) [13]. This project uses it for non-commercial academic research only and cites it as requested by its providers [3], [13]. No images are redistributed with this work. Derived files (split manifest and client partition) contain only identifiers and labels and will be shared under the same non-commercial terms with attribution.

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